ConvFormer-CD: Hybrid CNN–Transformer With Temporal Attention for Detecting Changes in Remote Sensing Imagery

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***Abstract*— Recently, the combination of Transformers and convolutional neural networks (CNNs) has witnessed significant advancements in change detection (CD) tasks. However, it remains unexplored how to interactively integrate long-range dependency and local information to enhance the model's global–local context awareness for effectively mitigating pseudo-changes. In addition, accurate identification and distinction of building changes from complex backgrounds still pose challenges due to the insufficient semantic context modeling across time between bi-temporal images. To address these issues, we propose a hybrid model ConvFormer-CD with parallel convolution and multihead self-attention (MSA). This combination enables better interaction of global and local information, thereby enhancing the adaptability to complex scenarios. Moreover, we introduce a novel module called Temporal Attention to establish cross-temporal semantic relationships between image pairs, effectively highlighting change regions by learning shared and nonshared semantics. This enables our model to accurately detect changed targets even in scenarios characterized by intricate geo-spatial arrangements and distributions. To further refine the differences in bi-temporal images, we propose a difference integration module (DIM) that connects the encoder and the decoder to fuse high-level semantic features across channels. We conduct extensive experiments on four benchmark datasets, including LEVIR-CD, LEVIR-CD+, WHU-CD, and S2Looking-CD, which demonstrates that the proposed ConvFormer-CD outperforms other state-of-the-art (SOTA) methods. Our codes will be available at https://github.com/taomi-lab/ConvFormer-CD.**

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I. Introduction

CHANGE detection (CD) is a critical remote sensing (RS) task that involves identifying and analyzing changes of interest between different temporal RS images of the Earth's surface. This technique has widespread applications in urban planning [1], environmental monitoring [2], and resource management [3], which plays a vital role in achieving sustainable development and promoting social and economic growth [4].

With the widespread availability of high-resolution RS images, the presence of pseudo-changes significantly impairs the accuracy of CD [5]. As shown in Fig. 1(a), this is primarily because the enhanced spatial resolution leads to richer ground details, where factors such as vehicle movements and cargo container placements can give rise to pseudo-changes. Additionally, buildings may exhibit uneven spectral characteristics under the influence of seasonal and lighting variations, thereby resulting in pseudo-change. Regarding these pseudo-changes, relying solely on local features from convolutional neural networks (CNNs) [6] or global features from Transformers [7] proves to be insufficient. This is because local features [8], [9] mainly focus on fine details, such as building boundaries, surface textures, and small objects, lack a contextual perception of the entire image, whereas global features [10], [11] emphasize the overall layout of roads and buildings from a broader, macroscopic perspective. Integrating local features extracted by convolution with global features derived from self-attention mechanisms helps mitigate pseudo-changes, enhances the detection method's adaptability to complex scenes, and improves the overall accuracy (OA) of CD tasks.

However, existing methods [12], [13] perform global modeling based on features obtained through convolution, which often treat the extraction of local and global features as relatively independent stages and usually operate at different levels. This will lead to insufficient information transfer and integration when processing local and global information, thereby compromising feature integrity and representation capability, which, in turn, affects the CD performance.

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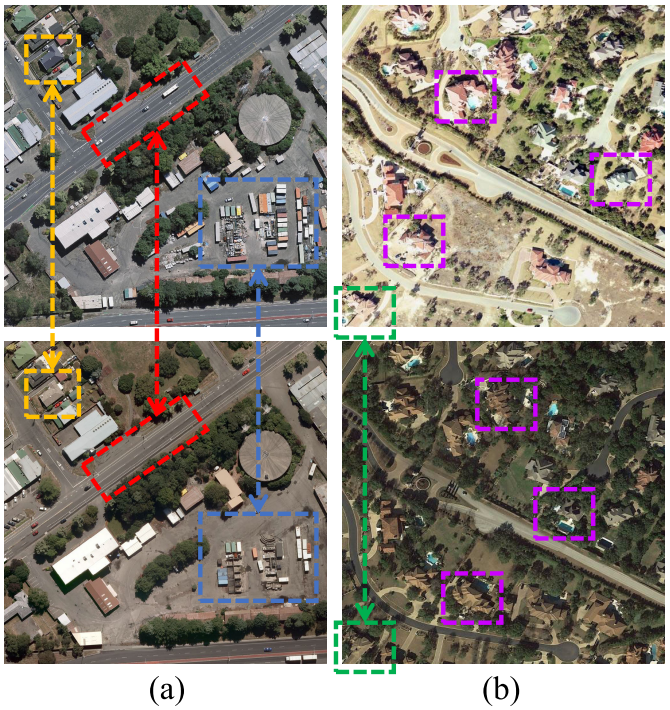


Fig. 1. Illustration of pseudo-changes and cluttered spatial distributions in CD. (a) Pseudo-changes are caused by the movement of containers (blue boxes) or vehicles (red boxes), and roof color alterations (yellow boxes). (b) The distribution and spatial arrangement of buildings in RS scenes are highly diverse (purple boxes). Variations in imaging conditions can lead to significant differences in images captured at different times in the same location (green boxes), making CD more challenging.

In contrast, our approach stands apart by performing parallel operations of convolution and the multiscale self-attention mechanism in a single block. This design integrates global and local features at the same layer, enabling the simultaneous extraction and processing of both feature types within the same stage. This can avoid the problem of information loss caused by the separation of feature extraction and processing stages in traditional methods. The feed-forward neural network (FFN) that comes with the Transformer block also helps global and local features combine better, hereby fully leveraging their advantages.

As shown in Fig. 1(b), in addition to the influence of various pseudo-changes, geospatial objects' distribution and spatial arrangement in RS images are typically highly complex. For instance, buildings in urban residential areas are usually densely packed, while in rural areas, the layout of buildings is more scattered without overall planning. Accurately identifying and distinguishing buildings from the background poses a significant challenge in such scenarios. Fortunately, as shown in Fig. 1(b), there is prior knowledge in CD indicating that buildings in dual-temporal images typically share some similarities in shape, structure, texture, or common background. Therefore, there is potential to use single-image information for object mining as well as establish the semantic correspondence between two images, which can leverage context from cross-temporal images to identify and discriminate a scene or object. In the CD task, these intertemporal semantic relationships reveal a strong coupling connection between pairs of dual-temporal images, where two temporal phases exhibit certain shared and unshared semantics that can be employed to distinguish changed components from unchanged ones.

Inspired by this, during feature extraction, we not only rely on the internal information in single-phase images but also delve into the interactions between the two-phase images. However, previous approaches such as Siamese networks [14] typically focus on extracting features independent of individual temporal phases in the encoder stage, neglecting the rich semantic context inherent in dual-temporal data. This limitation restricts the model's capacity to capture cross-image contextual information from the other related temporal phase, resulting in inaccurate CD results. Therefore, we design a "Temporal Attention" module that explores semantic correspondences and differences between dual-temporal images by cooperatively establishing cross-time dependencies through the temporal difference-guided attention mechanism. This module highlights change regions by learning homogeneous and heterogeneous features with co- and cross-attention mechanisms between the two temporal phases, enabling our model to better recognize and differentiate change targets, especially when the spatial arrangement and target distribution are extremely complex.

In this article, we propose a novel hybrid network named ConvFormer-CD that integrates the parallel convolution path and self-attention/temporal-attention path, avoiding the loss of information during the fusion process of global and local information, and enhancing representative difference features from bi-temporal images. We introduce a local enhancement module (LEM) based on the CNN, which runs in parallel with a self-attention mechanism. This integration enables the network to obtain fine-grained features and long-range correlation information, thereby reducing the impact of pseudo-changes caused by various factors in CD scenarios. We propose a Temporal Attention module that establishes semantic relationships across two temporal phases, utilizing shared common semantics and unshared semantics (i.e., uncommon components) between these phases to emphasize regions with changes. By interleaving calculations between self-attention and temporal attention, we can extract long-range correlations among geospatial objects (e.g., buildings) at different locations in a single temporal phase and also gather information about cross-temporal relationships of objects between image pairs. Furthermore, we replace the standard skip connection in the encoder–decoder architecture with a difference integration module (DIM) serving as the bridge, which integrates essential difference features learned from the encoder into the decoder, facilitating the capture of meaningful changes. In summary, ConvFormer-CD leverages rich contextual relationships by fully utilizing the difference information between the two temporal phases and the global–local information in a single temporal phase, ultimately improving model accuracy and suppressing pseudo-changes.

The main contributions of this article can be summarized as follows.

1) We propose a novel hybrid network with elaborate encoder blocks that integrate local convolutions and

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self-attention mechanisms in a parallel fashion. This unique combination enables our model to leverage the advantages of both CNN and Transformer architectures from a well-balanced interactive perspective, which can capture more comprehensive global–local information, facilitating effective extraction of differences between dual-temporal images.

2) We design a module called Temporal Attention, which can establish semantic relationships across the feature maps of image pairs by co-guiding the feature learning process of both temporal instances, making full use of the shared and unshared semantics between the two temporal images to discriminate the change components from the unchanged.
3) A plug-and-play module called the DIM was tailored for a CD. The module serves as a bridge to connect the encoder and the decoder, further enhancing the change information learned from the encoder. This approach strengthens the model's ability to discern meaningful changes while mitigating the impact of pseudo-changes.
4) We conducted extensive qualitative and quantitative experiments on four publicly available CD datasets. The results showcase the superior performance of our ConvFormer-CD model compared to existing state-of-the-art (SOTA) methods.

The rest of this article is organized as follows. Section II reviews the convolution- and Transformer-based CD methods. Our proposed method is described in detail in Section III, and experimental results are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. CNN-Based Methods

Due to the remarkable performance of CNNs in feature extraction, they have been widely utilized in various image-related tasks. Similarly, successful methods in natural scenes, such as SegNet [15] and UNet [16], have found extensive applications in RS CD tasks. In the context of CD, where dual-temporal images are involved, Zhan et al. [17] introduced Siamese networks [18], which enabled simultaneous processing of images from both temporal phases. Subsequently, this architecture became the foundation for many CD methods. Daudt et al. [9] introduced FC-EF, a fully convolutional CD network based on UNet. The FC-EF model was further improved by integrating Siamese architecture, forming two variants: FC-Siam-conc and FC-Siam-diff, which differ in their skip connections. In FC-Siam-conc, feature fusion between the encoder and the decoder is achieved through concatenation, while in FC-Siam-diff it is accomplished via differencing. Peng et al. [19] proposed an improved end-to-end CD network based on UNet++ [20]. This method utilizes dense skip connections in UNet++ to obtain multiscale semantic features. Furthermore, Peng et al. [19] proposed SNUNet-CD by associating UNet++ with the Siamese network. This approach emphasizes the significance of shallow-level information, as it contains finer details of objects, effectively alleviating the omission of small targets. Zhang et al. [21] proposed a difference-guided aggregation network that incorporates a weighted metric of change magnitude between bitemporal pixels to facilitate dual-temporal feature interaction and information fusion in complex scenarios.

Although these methods have demonstrated their capability in CD, they cannot often extract global features. The features obtained from these methods are typically sensitive to factors such as noises, angles, shadows, and context, resulting in limited robustness against pseudo-change [13]. To overcome these limitations, several improved algorithms have been proposed to encode image context in the spatiotemporal dimensions better and enhance feature representation. These approaches include techniques such as stacking additional convolutional layers [22], utilizing dilated convolutions [23], incorporating generative adversarial network (GAN) models [24], and employing attention mechanisms [25], [26], [27]. While these methods have made advancements, the limited receptive field of CNNs may impede the extraction of global contextual information from the input. Therefore, these methods still face challenges in fully harnessing global information and effectively capturing the meaningful differences between dual-temporal images to extract the regions of change components.

B. Transformer-Based Methods

The Transformer model [28] is initially proposed for machine translation tasks and it outperforms previous models based on recursion or CNNs [29]. Since then, Dosovitskiy et al. [7] introduced a Transformer into the computer vision field. The excellent performance of Transformer compared to CNN models in computer vision tasks has been reported by numerous studies.

Similarly, Transformer has achieved promising results in CD tasks [30]. Zhang et al. [31] combined a Swin Transformer with UNet, introducing a pure Transformer-based CD network called SwinSUNet. SwinSUNet is the first CD network exclusively built on the Transformer architecture, achieving superior results compared to previous methods based on pure convolution. Furthermore, Bandara and Patel [32] introduced a Transformer-based Siamese CD network called ChangeFormer. Utilizing a hierarchical multilayer perceptron (MLP) as the decoder, their approach surpasses many large-scale pure convolutional networks. However, the self-attention mechanism of the Transformer often fails to capture fine-grained local details. To address this limitation, Chen et al. [12] proposed a model called BiT. The BiT uses a Transformer encoder to globally enhance the features extracted by CNNs, which demonstrates that the hybrid model is more efficient than the pure convolutional networks. Li et al. [13] incorporated the Transformer and UNet within an encoder–decoder framework called TransUNetCD. Embedding the Transformer at the bottom of the UNet, the model aims to enhance and learn global information about semantic features extracted from the encoder. This implies that the combination of the Transformer and UNet can deliver superior performance.

However, BIT and TransUNetCD try to combine the Transformer and CNN in a sequential manner, which mainly uses the Transformer to perform global context modeling

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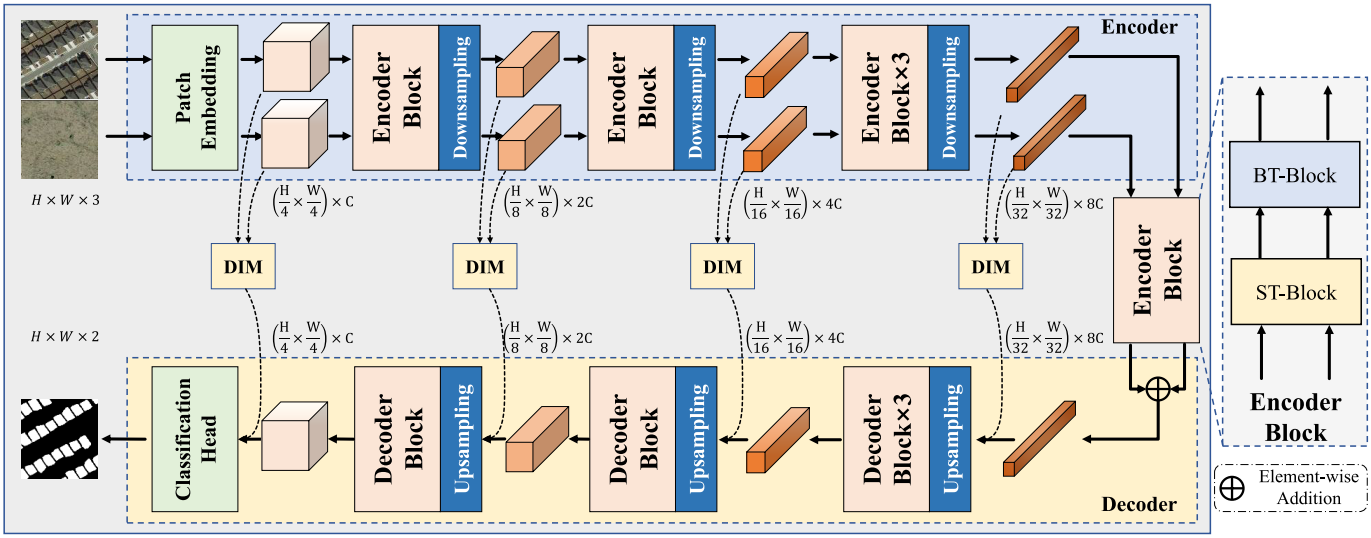


Fig. 2. Architecture of our proposed ConvFormer-CD.

of deep convolutional features. These approaches fall short in capturing comprehensive multiscale contextual information and do not fully exploit the combined potential of CNN and Transformer architectures. To address the above problem, we combine convolutional operations with self-attention mechanisms in parallel, which simultaneously extracts both local and global features with a synchronous interaction. In addition, our model embeds the parallel self-attention branch and convolution branch in each stage of the UNet framework to extract comprehensive global–local information across multiple scales, rather than performing CNNs followed by Transformers separately.

III. Methodology

A. Overall Architecture

Currently, some progress is being made in the combined usage of the Transformer and convolution for CD tasks. However, how to integrate long-range dependencies and local information in a fully interactive manner to enhance the local–global awareness of the model still needs further improvement. In addition, these methods face challenges in complex scenarios due to the lack of cross-temporal semantic contextual information modeling between image pairs. Motivated by these issues, a novel and more efficient CD model that combines convolution and Transformer is proposed in this article. Fig. 2 shows the overall architecture of the proposed ConvFormer-CD.

Specifically, the encoder utilizes a Siamese structure for concurrent feature extraction from bi-temporal images. Initially, the images undergo patch embedding to convert them into image tokens, which then progress through three consecutive stages with varying feature scales. Each encoder block incorporates a single-temporal local–global attention block (ST-Block) and a bi-temporal semantic difference attention block (BT-Block). These components are designed to extract global–local information from single-temporal images and learn semantic differences in dual-temporal images. The encoder and the decoder share a similar block architecture, but the decoder adopts a single-stream structure. Furthermore, we introduce a DIM module to bridge the encoder and the decoder, in contrast to the way of direct skip connection in UNet, to fuse the semantic difference information of dual-temporal features. Finally, the classification head uses the output of the decoder to generate the final change map.

B. Single-Temporal Local–Global Attention Block

Global and local features can work in concert to extract semantic and detailed information for CD tasks, improving accuracy detection. Nevertheless, current approaches typically stack convolutional and attentional layers alternately in series, making it impossible to capture both local and global dependencies at the same time without losing one kind of information. To address this problem, we propose the ST-Block to extract the single-phase image's global and local information in parallel.

The primary structure of the ST-Block is illustrated in Fig. 3(a), which mainly comprises two parallel branches: a convolution-based LEM and a window-based multihead self-attention (W-MSA) mechanism operation containing LayerNorm (LN) and MLP. For the input feature X_i, where i indices the two time phases, we input it into the ST-Block with the LEM to extract local information and the W-MSA mechanism to capture global information simultaneously. Unlike traditional multihead self-attention (MSA), W-MSA restricts the calculation of self-attention in a $D \times D$ local window rather than globally. The formula for self-attention is

$$Q = XW^q, \quad K = XW^k, \quad V = XW^v \tag{1}$$

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where X is the input for self-attention operations with dimensions $n \times d$. Additionally, we utilize three linear layers represented by weight matrices W^q, W^k, and W^v with dimensions $d \times d$. The query (Q), key (K), and value (V) matrices

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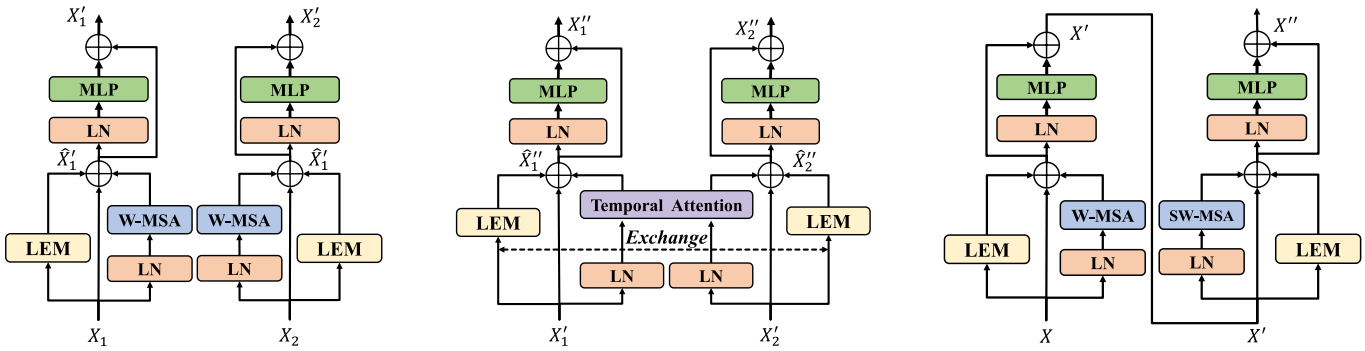


Fig. 3. Structures of the encoder and decoder blocks. The encoder stage is composed of (a) ST-Block and (b) BT-Block, which extract the single-temporal and dual-temporal features, respectively; the decoder stage includes the (c) Decoder Block for the prediction of CD.

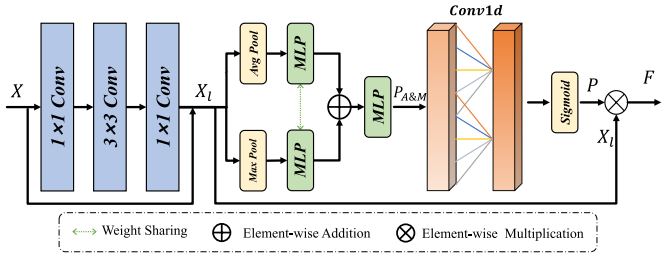


Fig. 4. Local enhanced module.

are also defined as $n \times d$ in size. Here, n refers to the resolution of Q, K, and V, while d corresponds to the channel dimension. The W-MSA branch employed in our ST-Block is designed to capture long-range dependencies in the images.

1) Local Enhancement Module: The structure of the LEM branch is shown in Fig. 4, where the input feature X_i undergoes a Residual convolution block, similar to the one utilized in ResNet, to extract local detailed information X_l

$$X_l = X + \text{Conv}_{1\times1}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(X))) \tag{3}$$

where $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ denote convolutional layers with 1×1 and 3×3 convolutional kernels, respectively. Then, we employ an approach akin to channel attention to mitigate the influence of extraneous channels, which may obscure key features and hinder the network from collecting fine-grained local semantic details. Specifically, we use two aggregation strategies (i.e., average and max-pooling) to obtain the channel dependencies $P_{A\&M}$. Subsequently, the learned dependencies $P_{A\&M}$ are passed through a 1-D convolutional layer with a kernel size of k, thereby further accentuating significant feature channels. To account for the variable number of channels for different layers, we have a mapping relation φ between k and input channel C

$$C = \varphi(k) = 2^{(\gamma \times k - b)} \tag{4}$$

where we set the controllable parameters γ and b to 2 and 1, respectively. We obtain the channel attention weight P after this 1-D convolution operation. The above process can be expressed as follows:

$$\begin{aligned} P_{A\&M} = \sigma(&\text{MLP}(\text{MLP}(\text{AvgPool}(X_l)) \\ &+ \text{MLP}(\text{MaxPool}(X_l)))) \end{aligned} \tag{5}$$

$$P = \sigma(\text{Conv1D}(P_{A\&M})) \tag{6}$$

where σ represents the sigmoid function and MLP represents the multilayer perceptron module [33].

Finally, we multiply the channel weight P with the output X_l of the Residual convolution block to get the refined local feature F

$$F = X_l \odot P. \tag{7}$$

In summary, the LEM not only extracts local information but also retains the original channel information while highlighting the significant channels relevant to changes. By running the LEM in parallel with the self-attention mechanism, the ST-Block achieves a balance between preserving important feature details and establishing long-term interdependence. The parallel design enables our model to realize multiple interactions of local and global features to prevent the loss of critical information and enhance the adaptability to complex scenes, which can be more effective in identifying areas of interest and mitigating the impact of pseudo-changes.

The final output of the ST-Block can be expressed as

$$\begin{aligned} \hat{X}_i' &= F_i + X_i + \text{W-MSA}(\text{LN}(X_i)) \\ X_i' &= \text{MLP}\big(\text{LN}\big(\hat{X}_i'\big)\big) + \hat{X}_i' \end{aligned} \tag{8}$$

where $i = 1, 2$ denotes two temporal phases and LN means the LayerNorm layer [33].

C. Bi-Temporal Semantic Difference Attention Block

It should be noted that the essence of RS CD is to analyze changes between the pairs of two images. In other words, the core of the CD task is to train detectors that can identify changes in images taken at different times but at the same spatial location. Existing CD models typically extract deep semantic features from each temporal phase separately and then fuse these features in the decoder, using skip connections to capture corresponding information between the two temporal images. Therefore, the encoder stage often fails to capture cross-image contextual information from the other related temporal phase, resulting in insufficient semantic correspondences and differences between dual-temporal images. To tackle the above problem, we propose the BT-Block, as shown in Fig. 3(b). The overall structure of BT-Blocks

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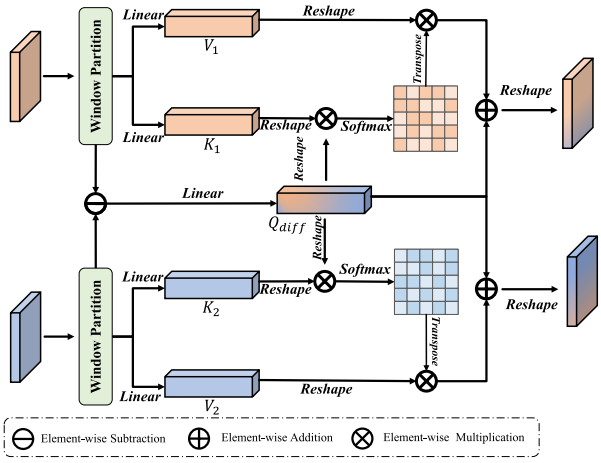


Fig. 5. Illustration of the temporal attention.

resembles that of ST-Blocks, with a focus on modeling cross-temporal interactive relationships in the BT-Blocks. We design a unique attention mechanism named Temporal Attention for feature-difference operations in the BT-Block, in place of the conventional self-attention mechanism in the ST-Block.

Temporal Attention: The structure of temporal attention is illustrated in Fig. 5. We divide the two temporal features into smaller windows via window splitting, enabling the computation of the elaborated temporal attention mechanism in each split window. There are underlying cross-temporal semantic relationships between pairs of dual-temporal images, which implies shared and unshared semantics of the changed and unchanged components. Hence, to filter out unchanged backgrounds and enhance changed targets, we subtract the features of the two temporal phases and then pass the differences through a linear layer to generate a difference query vector (Q_{diff}). Meanwhile, the features of each temporal phase are also passed through a linear layer to generate the key vector (K) and value vector (V) representations, respectively. Next, Q_{diff} is used to query the K_1 and K_2 vectors of the two time phases. The query results are multiplied with the V_1 and V_2 vectors of each time phase, respectively, to highlight the regions with difference information between the two temporal phases. Finally, to accentuate the difference information across image pairs, the output results are added to the Q_{diff} vector by a residual connection, yielding enhanced bi-temporal features with emphasis on the detected changes. This allows us to exploit cross-temporal dependencies through temporal attention to capture homogeneous and heterogeneous features between the two temporal phases. The specific calculation process can be expressed as follows:

$$K_i = X_i' W_i^k \tag{9}$$

$$V_i = X_i' W_i^v \tag{10}$$

$$Q_{\text{diff}} = \left(X_2' - X_1'\right) W^q \tag{11}$$

$$\text{SW-TA}\left(X_i'\right) = Q_{\text{diff}} + \text{SoftMax}\left(\frac{Q_{\text{diff}} K_i^T}{\sqrt{d_k}}\right) V_i \tag{12}$$

where $i = 1, 2$ denotes two temporal phases and X_i' denotes the image features of two temporal phases extracted by the ST-Block. K_i and V_i denote the key and value vectors of each temporal feature, respectively. Q_{diff} denotes the query vector of the difference between X_1' and X_2'.

Unlike standard MSA, the temporal attention mechanism incorporates co- and cross-attention calculations between bi-temporal image features to amplify the representation of changed regions while mitigating the impact of unchanged regions in complicated situations. The receptive field may be constrained by the $D \times D$ window in which the W-MSA operates in the ST-Block. The temporal attention in the BT-Block uses the shifted windowing scheme following the Swin transformer, denoted as shifted window temporal attention (SW-TA) in (12). By alternately applying the ST-Block and BT-Block, our model can increase the limited receptive field.

Besides global difference information, local difference information is also crucial for our task. Therefore, in the LEM branch of the BT-Block, we incorporate a feature exchange operation based on [34]. This operation consists of channel exchange and spatial exchange in two temporal phases, where channel exchange refers to exchanging features in the channel dimension and spatial exchange involves exchanging features in the spatial dimension. After the feature exchange operation, the exchanged features undergo processing in the LEM to extract the local difference information F_i^{ex}.

In summary, the BT-Block mainly explores semantic correspondences and differences between dual-temporal images through the temporal attention mechanism to capture more comprehensive information, so that our model can detect changes in objects in complex scenarios. The output of the BT-Block module can be expressed as

$$\hat{X}_i'' = F_i^{\text{ex}} + X_i' + \text{SW-TA}\left(\text{LN}\left(X_i'\right)\right) \tag{13}$$

$$X_i'' = \text{MLP}\left(\text{LN}\left(\hat{X}_i''\right)\right) + \hat{X}_i''. \tag{14}$$

D. Difference Integration Module

Buildings often exhibit varying spectral characteristics due to seasonal changes and different lighting conditions. Additionally, the acquired bi-temporal images may present pseudo-changes caused by some irrelevant objects, such as the movement of containers or vehicles, which disturbs the CD task. A reliable CD model aims to retain meaningful semantic changes while filtering out pseudo-changes. Hence, it is crucial to refine the information on semantic differences of dual-temporal image features when detecting changes. In dual-temporal image features, different channels can convey distinct information. Some channels primarily capture patterns of change and reflect differences between the images, while other channels may emphasize background-related information. To address this, we insert a DIM between the encoder and the decoder, which efficiently captures the semantic changes of the targets across channels.

The structure diagram of the DIM module is illustrated in Fig. 6. The input to the module is the difference feature X_d derived from the encoder between two temporal phases. First, difference feature maps X_d are separately pooled using average pooling (AvgPool) and maximum pooling (MaxPool)

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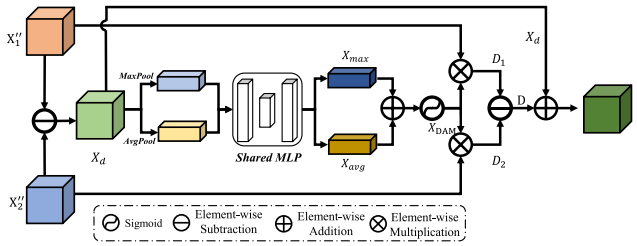


Fig. 6. Difference integration module.

operations. The pooled feature maps are then fed into two MLP layers, and the outputs are summed together. Then, the summed features pass through a Sigmoid function to obtain the difference attention map X_{DAM}. The above process can be formulated as

$$X_d = X_1'' - X_2'' \tag{15}$$

$$X_{\mathrm{avg}} = \mathrm{MLP}(\mathrm{ReLU}(\mathrm{MLP}(\mathrm{AvgPool}(X_d)))) \tag{16}$$

$$X_{\mathrm{max}} = \mathrm{MLP}(\mathrm{ReLU}(\mathrm{MLP}(\mathrm{MaxPool}(X_d)))) \tag{17}$$

$$X_{\mathrm{DAM}} = \sigma(X_{\mathrm{avg}} + X_{\mathrm{max}}) \tag{18}$$

where σ represents the sigmoid activation function and ReLU represents the ReLU activation function.

Finally, the differential attention map X_{DAM} is multiplied with the original input image feature X_i to activate the differential enhanced feature map D_i. Subsequently, we perform a subtraction between the enhanced difference features D_1 and D_2 to obtain the final difference-integrated feature D. To retain the original information, we add the difference-integrated feature D to the original differential feature X_d to obtain the final output X_{out}

$$D_i = X_{\mathrm{DAM}} \odot X_i'', \quad i = 1, 2 \tag{19}$$

$$X_{\mathrm{out}} = D_1 - D_2 + X_d. \tag{20}$$

E. Decoder

Our decoder is designed in a single-stream architecture. As shown in Fig. 2, similar to the encoder, our decoder comprises three stages. The first stage consists of three decoder blocks, while the remaining two stages each have one decoder block. The input for each decoder stage is formed by adding the fused difference features obtained from the DIM to the output of the previous decoder stage. The detailed structure of a decoder block is illustrated in Fig. 3(c), where the computation within each decoder block can be expressed as follows:

$$\hat{X}' = X + \mathrm{LEM}(X) + \mathrm{W\text{-}MSA}(\mathrm{LN}(X)) \tag{21}$$

$$X' = \mathrm{MLP}(\mathrm{LN}(\hat{X}')) + \hat{X}' \tag{22}$$

$$\hat{X}'' = X' + \mathrm{LEM}(X') + \mathrm{SW\text{-}MSA}(\mathrm{LN}(X')) \tag{23}$$

$$X'' = \mathrm{MLP}(\mathrm{LN}(\hat{X}'')) + \hat{X}''. \tag{24}$$

Finally, the output feature from the last decoder stage is fed into a classification head to generate the final change map. The classification head comprises a convolutional layer and two consecutive patch expansion modules. The change map is expanded to the size of the original input image through two successive patch expansion modules, and a 1×1 convolutional block is used for pixel-by-pixel binary classification to predict the final CD results, which can be expressed as

$$y = \mathrm{Conv}_{1\times1}(\mathrm{Expand} \times 2(X'')) \tag{25}$$

where Expand $\times$ 2 denotes two successive patch expansion modules.

F. Loss Function

We employ an end-to-end training strategy to optimize the proposed ConvFormer-CD model, wherein the prediction of each change map is supervised through a union of binary cross-entropy (BCE) loss and Dice loss. The total loss of the training is defined as follows:

$$L_{\mathrm{total}} = L_{\mathrm{bce}} + \alpha L_{\mathrm{dice}} \tag{26}$$

where L_{bce} denotes BCE loss, L_{dice} denotes Dice loss, and the parameter α is used to balance the contribution of the two losses and is empirically set to 0.4.

BCE is frequently employed in binary classification tasks to measure the discrepancy between predicted results and the true labels. BCE is highly appropriate for CD tasks, where pixels are classified as either changed or unchanged. The BCE loss can be defined as

$$L_{\mathrm{bce}}(y, p) = -\frac{1}{H \times W} \sum_{i=1}^{H \times W} (y_i \log(p_i) + (1 - y_i) \log(1 - p_i)) \tag{27}$$

where H and W denote the height and width of the image, respectively, y_i denotes the ground truth of pixel i, where $y_i = 1$ represents the changed category and $y_i = 0$ represents the unchanged category. p_i represents the predicted probabilities of the changed category and $1 - p_i$ vice versa.

In image segmentation tasks, a region-based loss function like Dice loss is proposed to address the issue of sample imbalance. The Dice Loss can be defined as

$$L_{\mathrm{dice}} = 1 - \frac{2|Y \cap P|}{|Y| + |P|} \tag{28}$$

where Y and P represent the sets of pixels in the ground truth and prediction maps, respectively. $|Y|$ and $|P|$ denote the number of pixels in the ground truth and prediction maps, respectively, while $|Y \cap P|$ represents the number of pixels in their intersection. The smaller the Dice Loss value, the higher the similarity between the prediction map and the ground truth. The incorporation of Dice loss aids the model in learning boundary information more effectively, leading to improved accuracy and completeness of the results.

IV. Experiments

A. Datasets

To evaluate the effectiveness of our proposed approach, we adopt four commonly used CD datasets: LEVIR-CD,

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LEVIR-CD+, WHU-CD, and S2Looking-CD. The details of each dataset are depicted as follows.

LEarning, VIsion, and Remote sensing Change Detection (LEVIR-CD) [35] is a large-scale change detection dataset with high resolution. It consists of 637 pairs of RS images captured from 2002 to 2018 in cities of Texas, USA. The spatial dimensions of the images are 1024 × 1024 with a resolution of 0.5 m/pixel. Notably, the LEVIR-CD dataset contains a diverse range of 31 333 individual change buildings, including cottage homes, tall apartments, small garages, and large warehouses. In our experiments, we follow the dataset partitioning scheme provided by the original authors, where the ratio of training, validation, and test sets is 7:1:2. To overcome the limited GPU memory capacity, according to [36], we employed a sliding window strategy to divide the original images into subimages of size 224 × 224 with a 64-pixel overlap, to get enough training samples.

LEVIR-CD+ [35] is constructed by augmenting the existing LEVIR-CD dataset with additional samples and corresponding labels. To facilitate the training and evaluation process, we adopt the dataset partitioning scheme provided by the original authors. And the original image is cropped to a nonoverlapping image of size 224 × 224.

The WHU Building Change Detection Dataset (WHU-CD) [37] is a dataset specifically designed for building change detection tasks. It consists of a pair of aerial images with a spatial resolution of 0.2 m/pixel and a spatial size of 32 507 × 15 354. These images show the building changes that took place in the Christchurch, New Zealand, region between 2011 and 2016 when a 6.3-magnitude earthquake struck. The dataset contains a total of 12 796 buildings, providing a comprehensive coverage of the area. As the dataset providers did not propose a formal partitioning scheme for the dataset, the original images were cropped into nonoverlapping images of size 224 × 224 in experiments. Subsequently, the dataset was randomly split into three subsets: training, validation, and testing. The numbers of these subsets are 8060, 1007, and 1007 image blocks, respectively.

The Satellite Side-Looking CD (S2Looking) dataset [38] is a collection of 5000 images obtained from three satellite sensors: Gaofen (GF), Super Vision (SV), and Beijing-2 (BJ-2). These images were captured between 2017 and 2020, with 3500 images designated for training and 1500 images for testing. The dataset covers a rural area worldwide, encompassing various complex scenes. Each image in the S2Looking dataset has a size of 1024 × 1024 pixels, and the satellite imaging provides a resolution ranging from 0.5 to 0.8 m. Compared to traditional top-down satellite images, the side-view perspective of the S2Looking dataset captures more detailed information about ground objects. This makes the dataset suitable for practical applications requiring a comprehensive and complex scene representation. To facilitate data processing and analysis, we divided each image into nonoverlapping blocks of size 224 × 224 pixels, resulting in a total of 100 000 images. Among these, 87 500 images were utilized for training, while the remaining 12 500 images were reserved for testing purposes.

TABLE I

COMPARISON OF PARAMETERS AND COMPUTATIONAL COSTS OF THE PROPOSED MODELS WITH VARYING CHANNEL SCALES

Model	Channle C	Params. (M)	Flops (G)
Convformer-CD/48	48	37.72	5.14
Convformer-CD/96	96	150.66	20.39

B. Evaluation Metrics

To assess the effectiveness of the methodology presented in this article, we employ the following six evaluation metrics: Precision (Pre), Recall (Rec), $F1$-score ($F1$), IoU, and OA. Their calculation formulas are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{29}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{30}$$

$$F_1 = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \tag{31}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{32}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{33}$$

where TP/TN stands for the number of correctly identified positive/negative samples, and FP/FN is the count of wrongly classified negative/positive samples.

C. Implementation Details

In our experimental trials, we employed PyTorch as the deep learning framework and we trained the proposed model on a 3090 graphics card with 24 GB memory. During the training phase, to improve the robustness of the model, similar to other CD models [42], [43], we use several common data augmentation techniques, including flipping, rescaling, cropping, and Gaussian blurring, all of which are applied during the training phase. Since the Swin Transformer backbone was trained from scratch without using any pretrained models, we trained more epochs compared to other models, with a total of 400 training epochs (max_epoch). Our model uses the Adam algorithm for optimization. The batch size is 32, the initial learning rate (lr) is 2e-3, and the learning rate decreases following the cosine schedule [44]. In addition, as shown in Fig. 2, after passing through the patch embedding layer, the input image is mapped to a feature map with a channel dimension of C. By setting the value of C, we obtain two models with different channel dimensions: Convformer CD/48 and Convformer CD/96, corresponding to $C = 48$ and $C = 96$, respectively. Due to the varying channel dimensions of the hidden layers, the number of parameters (Params.) and floating-point operations (FLOPs) of the model also differ, as detailed in Table I.

D. Comparison Experiments

In the comparative experiments, we compared the performance with some deep learning-based methods, including the

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TABLE II

QUANTITATIVE COMPARISON RESULTS ON THE LEVIR-CD AND LEVIR-CD+ DATASETS. THE CONVENTION OF COLOR AND FONT WEIGHT: **THE BEST IN BOLD RED** AND **THE SECOND-BEST IN BOLD BLACK**

Method	**Params. (M)**	**Flops (G)**	LEVIR-CD					LEVIR-CD+				
			Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
FC-EF$_{18}$ [9]	1.35	3.57	82.27	66.28	73.41	58.00	97.55	70.60	73.64	72.09	56.36	97.68
FC-Siam-conc$_{18}$ [9]	1.55	5.32	86.81	67.66	76.05	61.36	97.83	78.07	68.22	72.82	57.25	97.92
FC-Siam-diff$_{18}$ [9]	1.35	4.73	86.55	74.38	80.00	66.68	98.11	79.94	72.55	76.06	61.37	98.14
BiT$_{22}$ [12]	3.04	7.66	91.91	88.26	90.04	81.89	99.01	83.62	79.01	81.25	68.42	98.51
SNUNet$_{22}$ [39]	12.03	47.98	90.82	89.59	90.20	82.16	99.01	80.52	79.43	80.82	68.17	98.42
ChangeFormer$_{22}$ [32]	41.03	202.79	92.05	88.80	90.40	82.48	99.04	77.95	78.82	78.38	64.45	98.22
ICIFNet$_{22}$ [40]	23.84	22.23	91.32	88.64	89.96	81.75	98.99	**87.79**	80.88	83.65	71.89	98.73
DMINet$_{23}$ [41]	6.24	12.72	**92.60**	89.39	90.96	83.43	99.09	83.85	83.67	83.75	72.05	98.67
Convformer-CD/48 (ours)	37.72	5.14	92.37	**90.52**	**91.44**	**84.23**	**99.13**	85.15	**84.30**	**84.72**	**73.49**	**98.76**
Convformer-CD/96 (ours)	150.66	20.39	**92.52**	**91.04**	**91.78**	**84.80**	**99.17**	**86.29**	**84.33**	**85.30**	**74.37**	**98.81**

TABLE III

QUANTITATIVE COMPARISON RESULTS ON THE WHU-CD AND S2LOOKING-CD DATASETS. THE CONVENTION OF COLOR AND FONT WEIGHT: **THE BEST IN BOLD RED** AND **THE SECOND-BEST IN BOLD BLACK**

Method	**Params. (M)**	**Flops (G)**	WHU-CD					S2Looking-CD				
			Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
FC-EF$_{18}$ [9]	1.35	3.57	78.79	68.15	73.08	57.58	97.61	45.02	50.28	47.51	31.15	98.67
FC-Siam-conc$_{18}$ [9]	1.55	5.32	75.89	69.30	72.61	56.99	97.51	60.60	59.88	60.24	43.10	99.05
FC-Siam-diff$_{18}$ [9]	1.35	4.73	80.44	54.59	65.04	48.19	97.20	60.00	59.63	59.81	42.67	99.04
BiT$_{22}$ [12]	3.04	7.66	84.78	86.52	85.64	74.89	98.59	70.42	59.71	64.63	47.74	99.13
SNUNet$_{22}$ [39]	12.03	47.98	85.58	80.99	83.22	71.26	98.44	69.73	57.30	62.91	45.88	99.10
ChangeFormer$_{22}$ [32]	41.03	202.79	91.83	88.02	89.88	81.63	99.12	74.32	60.81	66.89	50.25	99.20
ICIFNet$_{22}$ [40]	23.84	22.23	89.57	**91.82**	90.68	82.96	99.10	69.39	**65.25**	67.26	50.67	99.15
DMINet$_{23}$ [41]	6.24	12.72	89.49	**90.57**	90.02	81.86	99.02	**80.01**	56.80	66.44	49.74	**99.23**
Convformer-CD/48 (ours)	37.72	5.14	**95.14**	89.26	**92.13**	**85.41**	**99.26**	**75.53**	61.00	**67.49**	**50.94**	99.21
Convformer-CD/96 (ours)	150.66	20.39	**95.11**	89.49	**92.21**	**85.54**	**99.27**	74.12	**62.81**	**68.00**	**51.51**	**99.21**

classical CD methods FC-EF [9], FC-Siam Diff [9], and FC-Siam Conc [9], as well as four SOTA methods, including SNUNet [39], BiT [12], ChangeFormer [32], ICIFNet [40], and DMINet [41]. To ensure fairness in comparison with other methods, we reimplemented all the CD networks using publicly available code in the same experimental environment, including using the same data preprocessing method. Additionally, both the input and output data were set to a fixed size of 224 × 224 pixels to maintain consistency across all models. During the training process, we particularly focused on the $F1$ score, as it comprehensively considers both precision and recall and is thus more reflective of model performance. Therefore, we optimized the $F1$ score on the validation subset to select the most suitable hyperparameters for each compared method. The quantitative results for the four datasets are shown in Tables II and III. Figs. 7–9 demonstrate qualitative results of SNUNet, BiT, ChangeFormer, ICIFNet, DMINet, and our ConvFormer-CD. We do not provide the predicted change maps for FC-EF, FC-Siam-conc, and FC-Siam-diff, as these methods yielded fewer competitive visual results in experiments.

1) Parameter Comparison: The number of parameters and computational complexity of different models were compared in this study. Table II presents the parameters and floating-point operations for each model. The high number of parameters in our model compared to other models is due to the use of transformers as the backbone network. However, the computational complexity of our model is lower because we use a similar in-window computation of self-attention as the Swin Transformer. The FLOPs for our Convformer-CD/48 were lower than those for all models other than FC-EF, accounting for only 67% of the BiT model.

2) Experimental Results on the LEVIR-CD Dataset: Table II presents the results of various methods on the LEVIR-CD and LEVIR-CD+ datasets. CNN-based methods FC-EF, FC-Siam-conc, and FC-Siam-diff have the advantage of occupying smaller memory space but achieve the poorest results across all metrics. The proposed ConvFormer-CD outperforms other methods in terms of $F1$ score and IoU. However, DMINet achieves the highest precision but with a lower recall. Our method, ConvFormer-CD, exhibits excellent performance on the LEVIR-CD dataset, achieving a recall of 91.04% and an $F1$ score of 91.78%. The $F1$ metrics and IoU metrics of ConvFormer-CD/96 surpass those of the second-ranked DMINet by 0.82 and 1.37 points, respectively. ConvFormer-CD/48 achieves a higher $F1$ score and IoU than the second-ranked DMINet with margins of 0.48 points and 0.8 points while only requiring 40% floating-point operations of DMINet. This means that ConvFormer-CD/48 offers faster inference speed while still delivering impressive performance.

Fig. 7 illustrates the visualization results of our model on the LEVIR-CD dataset. As shown by the yellow circles in Fig. 7(a) and (b), our model excels at identifying both large and densely distributed small-scale buildings, especially

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noticeable in scenarios with partially obscured buildings near image boundaries. Moreover, as indicated by the yellow circles in Fig. 7(c) and (d), the complexity of the surroundings significantly increases the difficulty of accurately distinguishing between false changes and meaningful changes of interest in bitemporal images. Other SOTA methods incorrectly identified changed areas. In contrast, our proposed model demonstrates exceptional performance in detecting building changes within complex and cluttered backgrounds, accurately representing true semantic changes. The experimental results suggest that, due to the integration of the ST Block and BT Block, our model exhibits superior capability in detecting targeted changes in complex environments while effectively filtering out false positives.

3) Experimental Results on the LEVIR-CD+ Dataset: In Table II, it can be observed that FC-EF, FC-Siam conc, and FC-Siam diff do not outperform other methods. ICIFNet and DMINet achieve comparable $F1$ scores (within a difference of 0.1), with ICIFNet having the highest precision among all methods at 87.79%. However, its recall rate is much lower at 80.88. DMINet also demonstrates impressive performance in terms of $F1$ score, ranking second only to our model. Convformer-CD/96 maintains the best performance on the LEVIR-CD+ dataset, with an excellent recall rate of 84.33% and the highest $F1$ score of 85.30%. In addition, Convformer-CD/48 also achieves impressive results compared to other methods, with an $F1$ score improvement of nearly 1 point and an IoU improvement of approximately 1.5 points.

4) Experimental Results on the WHU-CD Dataset: Table III presents the results of various methods on the WHU-CD and S2Looking-CD datasets, where FC-EF, FC-Siam conc, and FC-Siam diff still have lower performance compared to other methods. We also noticed that SNUNet and BIT, which achieved high evaluation scores on the other datasets, failed to produce satisfactory performance on the WHU-CD dataset. Low precision means that pseudo-changes cannot be distinguished by these models. This can be attributed to their sensitivity to dataset size, as training on smaller WHU datasets increases the risk of overfitting. Likewise, while the proposed ConvFormer-CD method excels in the $F1$ score, with ConvFormer-CD/48 surpassing other methods by at least 1.45 points, ConvFormer-CD/96 performs comparably to ConvFormer-CD/48, likely due to the impact of overfitting caused by the larger number of model parameters.

Fig. 8 illustrates the visualization results of our model on the WHU-CD dataset. Due to the high resolution of the WHU-CD dataset, which reveals more geospatial details prone to causing pseudo-changes, mitigating false detections is a challenging task. From Fig. 8(a) and (b), similar to the visualization of the LEVIR dataset, it is evident that our model can accurately detect changes even when buildings appear near the image borders, ensuring complete representation. As shown by the yellow circles in Fig. 8(c), our model can distinguish false changes caused by the placement of containers better than other SOTA models. Moreover, for pseudo-changes caused by differences in cement or asphalt surfaces that resemble buildings [e.g., the yellow circle in Fig. 8(d)], our model effectively identifies them while capturing discriminative feature details. Overall, compared to other competitors, ConvFormer-CD demonstrates lower false positive rates and significantly fewer omissions in visual performance.

5) Experimental Results on the S2Looking-CD Dataset: After analyzing the results in Table III, it can be observed that the FC-EF, FC-Siam conc, and FC-Siam diff methods still perform poorly in terms of performance. SNUet and BIT did not exhibit good performance either. ICIFNet and DMINet both achieved good results, with DMINet obtaining the highest precision of 80.01, but with a lower recall of 56.80. In contrast, ICIFNet achieved the highest recall of 65.25%, but with a precision of only 69.39%. Our Convformer-CD/96 demonstrated superior performance on the S2Looking dataset, achieving a balanced precision and recall with a recall rate of 62.81%, precision of 74.12%, the highest $F1$ score of 68.00%, and an IoU of 51.51%. Furthermore, Convformer-CD/48 obtained commendable results compared to other counterpart methods, with an $F1$ score of 67.49% and an IoU of 50.94%.

Fig. 9 illustrates the visualization results of our model on the S2Lookintg-CD dataset. Compared to traditional top-down satellite imagery, the oblique perspective of the S2Looking dataset captures more detailed information about ground objects. However, this also poses greater challenges for change detection tasks, making it more difficult for models to distinguish change areas. Therefore, the accurate identification of changed buildings under this dataset highlights the superiority of our method in extracting changes in buildings within complex environments. Based on the visualization results, our approach has stronger robustness with a higher level of accuracy in detecting building changes compared to other methods. This more clearly demonstrates that leveraging semantic information across temporal phases can be beneficial in detection.

E. Ablation Studies

To independently evaluate the contribution of each component in our proposed network, we performed ablation experiments on the ConvFormer-CD model. The first row of Table IV is the baseline model, which is a Siamese U-shaped network with a vanilla Swin Transformer as the backbone. We validate the effectiveness of each module by incorporating our proposed modules into the baseline model. In this section, we use the LEVIR-CD and WHU-CD datasets to evaluate our ablated models, and the evaluation metrics are the same as in the comparison experiments. The ablation experiments were performed on the Convformer-CD/48 model. To maintain consistency in the experimental setup, all networks involved in the ablation studies adhere to the same implementation details as described in Section IV-C.

Table IV presents the results of the module's ablation experiments. From Table IV, it can be observed that when the baseline model is equipped with the proposed temporal attention module, there is a significant improvement in both the $F1$ score and IoU metric. The temporal attention module leverages the difference information between dual-temporal features to accurately locate the changed regions in the images. As a result, in the ablation experiments, the baseline model

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TABLE IV
ABLATION STUDIES OF THE PROPOSED MODULES ON THE LEVIR-CD AND WHU-CD DATASETS

Temporal Attention	LEM	DIM	**Params. (M)**	**Flops (G)**	LEVIR-CD					WHU-CD				
					Pre	Rec	F1	IOU	OA	Pre	Rec	F1	IOU	OA
			25.56	3.26	90.03	89.83	89.93	81.71	99.16	91.33	88.41	89.85	81.57	99.06
✓			25.29	3.21	91.19	90.07	90.62	82.86	99.05	94.04	87.52	90.66	82.92	99.12
	✓		37.99	5.18	92.30	89.83	91.05	83.56	99.09	94.10	89.17	91.52	84.37	99.20
		✓	25.57	3.26	91.52	89.33	90.41	82.50	99.21	92.02	88.80	90.38	82.45	99.08
	✓	✓	37.99	5.19	91.60	**90.92**	91.26	83.92	99.53	**95.36**	88.67	91.89	85.00	99.24
✓		✓	36.55	4.10	91.51	89.27	90.38	82.44	99.03	93.95	86.45	90.04	81.89	99.07
✓	✓		37.72	5.14	92.09	90.34	91.21	83.84	**99.53**	94.70	88.90	91.71	84.68	99.22
✓	✓	✓	37.72	5.14	**92.37**	90.52	**91.44**	**84.23**	99.13	95.11	**89.49**	**92.21**	**85.54**	**99.27**

TABLE V
EVALUATION OF DIFFERENT ATTENTION MECHANISMS IN CONVFORMER-CD

	Params. (M)	**Flops (G)**	LEVIR					WHU-CD				
			Pre	Rec	F1	IOU	OA	Pre	Rec	F1	IOU	OA
Self Attention [28]	37.99	5.19	91.60	90.92	91.26	83.92	**99.53**	**95.36**	88.67	91.89	85.00	99.24
Cross Attention [40]	37.99	5.19	92.63	89.75	91.17	83.79	99.11	95.36	88.54	91.82	84.88	99.23
Joint Attention [41]	37.99	5.19	**92.68**	89.97	91.30	84.00	99.12	95.01	89.09	91.95	85.11	99.24
Temporal Attention (ours)	37.72	5.14	92.37	**90.52**	**91.44**	**84.23**	99.13	95.11	**89.49**	**92.21**	**85.54**	**99.27**

TABLE VI
ABLATION STUDIES OF THE LOSS ON THE LEVIR-CD AND WHU-CD DATASETS

Loss function	LEVIR-CD					WHU-CD				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
BCE	**93.16**	89.45	91.26	83.93	99.12	**95.68**	88.37	91.88	84.97	99.24
Dice	92.13	90.42	91.27	83.94	99.12	94.25	88.97	91.53	84.39	99.20
BCE+Dice	92.37	**90.52**	**91.44**	**84.23**	**99.13**	95.14	**89.26**	**92.13**	**85.41**	**99.26**

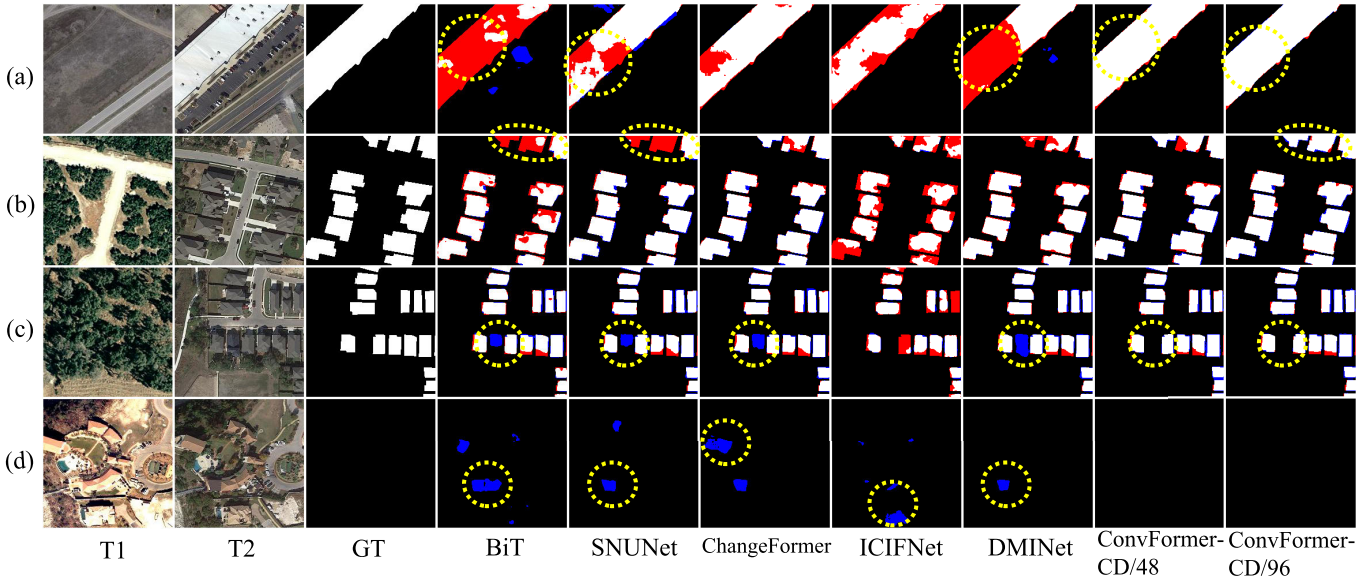


Fig. 7. Qualitative results of the different methods on the LEVIR-CD dataset. (a)–(d) Prediction results of all the compared methods for different bi-temporal image pairs, respectively. We highlight the TP areas in white, the FP areas in blue, and the FN areas in red. The black color denotes the TN areas.

with temporal attention outperforms the baseline, particularly in terms of precision, with improvements of 1.16 and 2.71 points on the LEVIR-CD and WHU-CD datasets, respectively. Regarding the LEM, it focuses on extracting local details using convolutions and emphasizes important and representative channels through channel-wise encoding. This enables LEM to capture more detailed information and enhance the accuracy of change detection. The results of the ablation experiments in Table IV also validate this observation. When the baseline model is augmented with LEM, there is a noticeable improvement in precision, with increases of 2.27 and 2.77 points on the LEVIR-CD and WHU-CD

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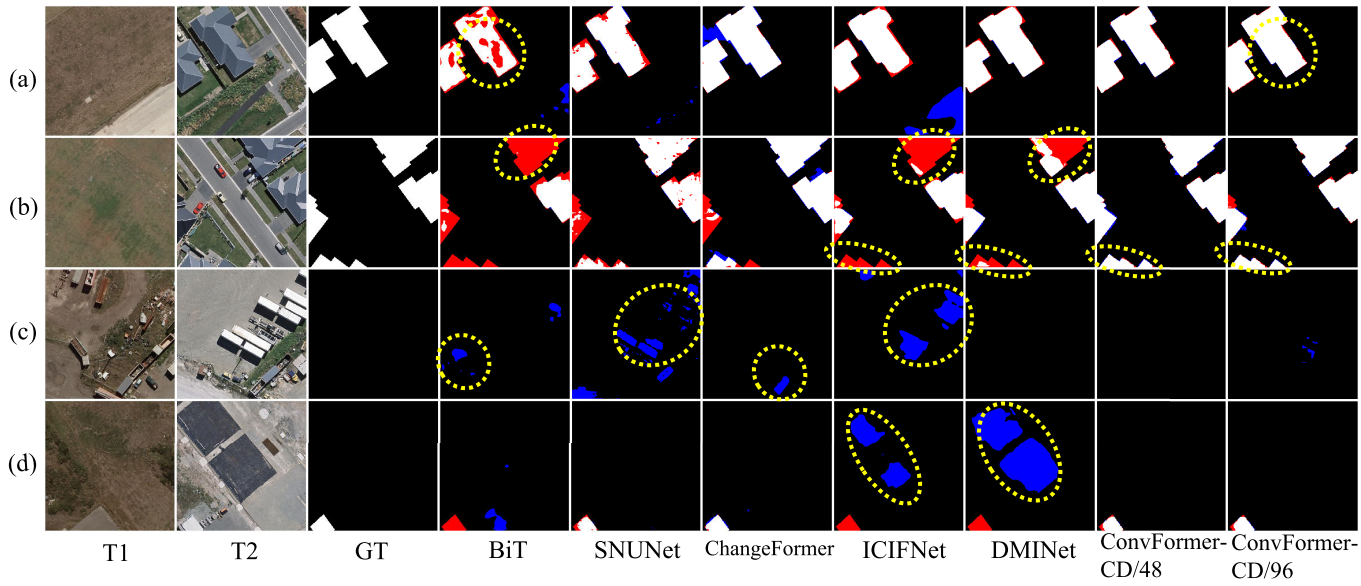


Fig. 8. Qualitative results of the different methods on the WHU-CD dataset. (a)–(d) Prediction results of all the compared methods for different bi-temporal image pairs, respectively. We highlight the TP areas in white, the FP areas in blue, and the FN areas in red. The black color denotes the TN areas.

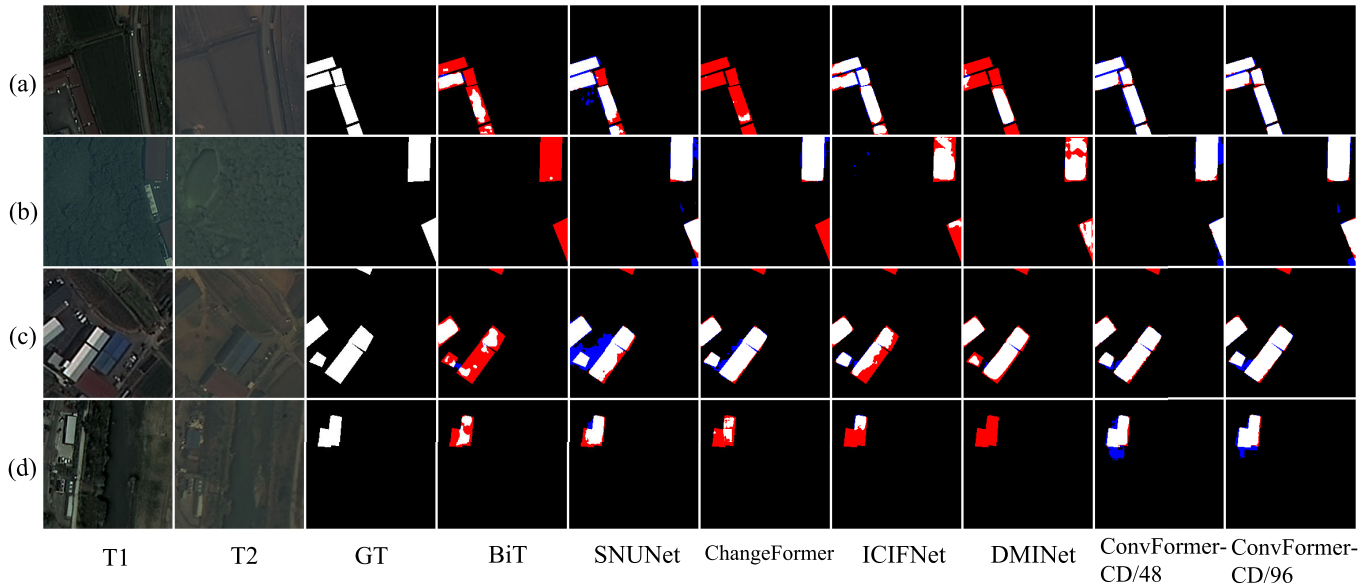


Fig. 9. Qualitative results of the different methods on the S2looking-CD dataset. (a)–(d) Prediction results of all the compared methods for different samples, respectively. We highlight the TP areas in white, the FP areas in blue, and the FN areas in red. The black color denotes the TN areas.

datasets, respectively. DIM connects the encoder and the decoder, uses the dual time-phase feature maps for difference operations, applies pooling operations to highlight channels with difference information, and fuses the difference-enhanced feature maps. In the ablation experiments, when the baseline model is augmented with DIM, there are notable improvements in precision (1.49 and 0.69 points) and $F1$ score (0.48 and 0.53 points) on the LEVIR-CD and WHU-CD datasets, respectively. Conversely, when DIM is removed from our final model, there is a decrease in all metrics, with a decrease of 0.23 and 0.5 points in the $F1$ score on the two datasets, respectively. Fig. 10 provides a visualization of the module ablation experiments. From the visual results (a), (b), and (g), it is evident that Temporal Attention has a positive impact on locating change regions in complex backgrounds. Furthermore, the visualizations in (a), (c), and (g) reveal that LEM extracts more detailed information, reducing pseudo-changes to some extent while preserving the areas of actual change. Additionally, the results in (a), (d), and (e) demonstrate that DIM effectively highlights differential information and enhances differential feature maps. In summary, the ablation experiments confirm the effectiveness of the proposed modules.

In addition to the module ablation experiments, we also analyzed the effectiveness of the designed attention mechanism of the temporal attention module. DMINet introduces a mechanism called joint attention, which concatenates the query vectors generated from the two temporal phases and uses the

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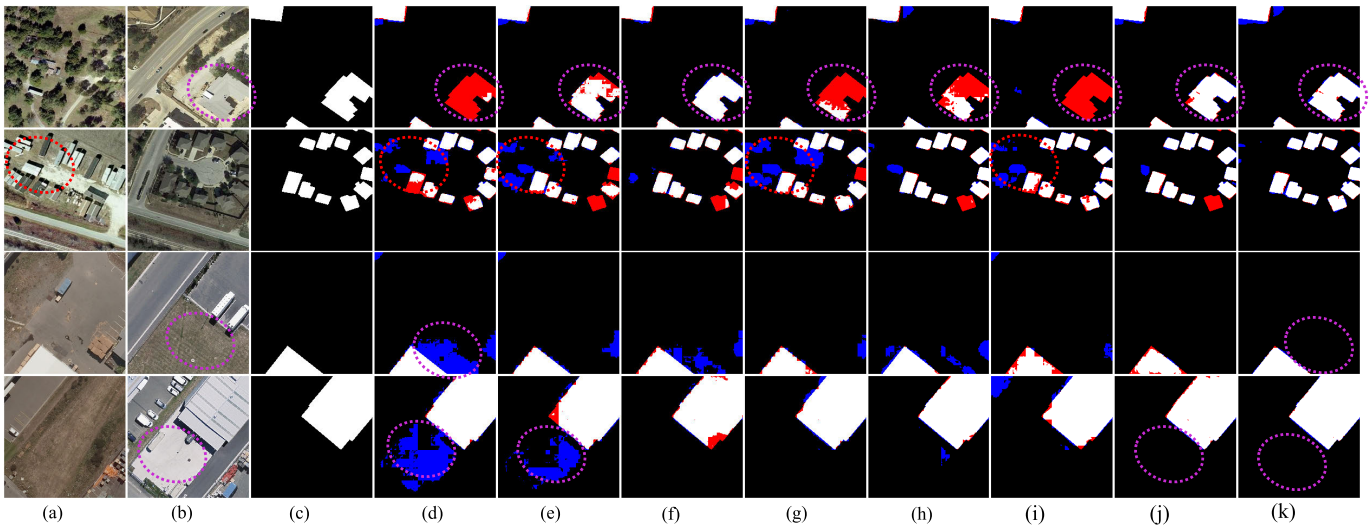


Fig. 10. Comparisons of different ablated models on the LEVIR-CD and WHU-CD datasets. From left to right, (a) T1, (b) T2, (c) GT, (d) baseline model, (e) only Temporal Attention, (f) only LEM, (g) only DIM, (h) LEM with DIM, (i) Temporal Attention with DIM, (j) Temporal Attention with LEM, and (k) our ConvFormer-CD/48. Colors: white for TP (i.e., "changed"), black for TN (i.e., "unchanged"), blue for FP, and red for FN.

concatenated vector to query the key (K) and value (V) vectors of each temporal phase. The advantage of this approach is that it better captures the shared features between the two temporal phases when extracting dual-temporal features. However, the essence of change detection is to identify the changed regions between the two images. Therefore, we propose the temporal attention module, which utilizes the difference features of the dual-temporal features to generate a difference query vector Q_{diff} and uses Q_{diff} to query the difference features in the dual-temporal features. We believe that this design can better capture the meaningful semantic differences in the dual-temporal images. Table V presents the results under different attention mechanisms, where we replace our temporal attention module with self-attention, cross-attention, and joint attention. We can observe from Table V that different attention mechanisms perform well on the training samples, but joint attention outperforms cross-attention and self-attention, with only a slight difference in precision but an improvement in recall compared to cross-attention. This indicates that joint attention plays a role in capturing the shared features between the dual-temporal phases. However, joint attention does not utilize such shared semantic features for difference extraction. Therefore, compared to other attention mechanisms like joint attention, temporal attention achieves the best $F1$ score, with results of 91.44% and 92.21% on the LEVIR-CD and WHU-CD datasets, respectively.

Finally, we conducted ablation experiments, comparing different loss functions. As shown in Table VI, combining Dice loss with BCE loss significantly enhances recall compared to using BCE loss alone, indicating a reduction in missed detections. This is attributed to Dice loss effectively dealing with imbalanced data in change detection, which stabilizes model training and enhances overall performance.

V. Conclusion

In this article, we propose a hybrid CNN–Transformer change detection network ConvFormer-CD. The network combines convolution and self-attention in a parallel manner and can effectively highlight change regions by establishing cross-temporal semantic relationships between pairs of bi-temporal images. Specifically, the proposed ST-Block operates convolution-based LEM in parallel with self-attention, which allows our model to strike a balance between preserving important feature details and establishing long-range inter-dependence. The BT-Block emphasizes regions of change by using temporal attention to learn homogenous and heterogeneous features, capturing semantic correspondences and differences between the two temporal phases. DIM is primarily responsible for acting as a bridge between the encoder and the decoder, integrating the essential difference information obtained from the encoder into the decoder. Our ConvFormer-CD achieves SOTA performance on four public datasets, and we verify the validity of each module.

However, ConvFormer-CD also exhibits certain limitations. It requires a sufficient number of training samples to learn robust deep feature representations, typically relying on large-scale, well-annotated change detection datasets. In practical scenarios, however, it is difficult to acquire a large volume of labeled samples. In addition, given that we utilize the Transformer, the ConvFormer-CD has a relatively large number of parameters, which poses difficulties in the case of limited computing resources. Thus, future work should focus on leveraging unlabeled samples to develop a lightweight semi-supervised or unsupervised ConvFormer-CD algorithm, aiming to enhance model scalability while reducing labeling costs.

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